For as long as we've known that (1) the universe started with a Big Bang and (2) it is currently expanding, the logical next question has been whether it will turn around and come back on itself, ending in a catastrophic Big Crunch. Starting with some very basic and reasonable physics assumptions, there appear to be only three possibilities for the future of an expanding universe, and they are all fairly direct analogs to what can happen to a ball thrown into the air.

You're standing outside, on the planet Earth. You throw a baseball straight up. You have an inhumanly good arm, just for the sake of argument. (And air resistance isn't a thing.) What will happen?

In the usual case, the ball goes up for a while, responding to that initial push you've given it, but it starts being slowed in its ascent by the gravitational pull of the Earth as soon as it leaves your hand.[fn5] Eventually it slows so much that it stops dead in the air and reverses course, falling back toward you and the planet you're standing on. But if you were to throw the ball incredibly fast—specifically, 11.2 km/s, the escape velocity of the Earth—you could in principle give the ball so much of a push that it leaves the Earth entirely, slowing down slightly all the while, and only comes to rest infinitely far in the future (or, I suppose, when it hits something else). If you throw it even faster, it'll be completely unbound from the Earth and just coast away forever.

The physics of an expanding universe follows very similar principles. There's the initial push (the Big Bang) that set off the expansion, and from that point onward the gravity of all the stuff in the universe (galaxies, stars, black holes, etc.) works against the expansion, trying to slow it down and pull everything back together again. Gravity is a very weak force—the weakest of all the forces of nature—but it's also infinite in range and always attractive, so even distant galaxies must pull toward each other. As in the baseball example, the question boils down to whether or not the initial push was enough to counteract all that gravity. We don't even have to know what the initial push was; if we measure the expansion speed now, and also measure the amount of matter in the universe, we can determine whether its gravity will be enough to make the expansion eventually stop. Alternatively, if we can infer the expansion speed in the distant past, we can determine how the expansion is evolving over time by comparing that number against the expansion speed today.[fn6]